

Proof of (i) $\implies$ (iii): Let $\{e_1, \dots, e_n\}$ be an orthonormal basis for V . Then

$$\langle U(e_i), U(e_j) \rangle = \langle e_i, e_j \rangle = 0$$

and

$$\|U(e_i)\| = \|e_i\| = 1.$$

So U maps an orthonormal set to another orthonormal set. Since U is bijective, it also maps linearly independent sets to linearly independent sets. The desired result follows. ■

Proof of (iii) $\implies$ (i): Fix any orthonormal basis $\{e_1, \dots, e_n\}$ for V . Let $v, w \in V$ be arbitrary vectors such that

$$v = \sum_{i=1}^n \alpha_i e_i \quad \text{and} \quad w = \sum_{j=1}^n \beta_j e_j.$$

Hence

$$U(v) = \sum_{i=1}^n \alpha_i U(e_i) \quad \text{and} \quad U(w) = \sum_{j=1}^n \beta_j U(e_j).$$

Now $\{U(e_1), \dots, U(e_n)\}$ is an orthonormal basis for V , meaning

$$\begin{aligned} \langle U(v), U(w) \rangle &= \left\langle \sum_{i=1}^n \alpha_i U(e_i), \sum_{j=1}^n \beta_j U(e_j) \right\rangle \\ &= \sum_{i=1}^n \alpha_i \overline{\beta_i} = \langle v, w \rangle. \end{aligned}$$
■

Proof of (iv) $\implies$ (ii): Let $x \in V$ be arbitrary. Then

$$\langle x, x \rangle = \|x\|^2 = \|U(x)\|^2 = \langle U(x), U(x) \rangle = \langle x, U^* U(x) \rangle.$$

So

$$\langle x, x - U^* U(x) \rangle = 0 \implies \langle x, (I - U^* U)(x) \rangle = 0.$$

Let $S = I - U^* U$. Note S is self-adjoint since

$$S^* = I^* - (U^* U)^* = I - U^* U = S.$$

By the Spectral Theorem for self-adjoint operators, there is an orthonormal basis β for V consisting of eigenvectors of S . Let $\beta = \{v_1, \dots, v_n\}$ where

$$Sv_i = \lambda_i v_i.$$

Now $\langle x, Sx \rangle = 0$ for any $x \in V$. Setting x to v_i , we get

$$\begin{aligned}\langle v_i, Sv_i \rangle &= \langle v_i, \lambda_i v_i \rangle \\ &= \overline{\lambda_i} \langle v_i, v_i \rangle\end{aligned}$$

Since $\|v_i\|^2 > 0$, we have $\overline{\lambda_i} = 0$. Hence every eigenvalue $\lambda_i = 0$. Thus $S = T_0$. Hence

$$S = I - U^*U = T_0 \implies I = U^*U. \quad \blacksquare$$

Theorem

Spectral Theorem for Unitary Operators

Let T be a linear operator on a finite-dimensional real inner product space V . Then V has an orthonormal basis consisting of eigenvectors of T , and the corresponding eigenvalues have absolute values of 1 if and only if T is self-adjoint and unitary (or Hermitian and orthogonal if V is a vector space over $\mathbb{R}$).

Proof: Suppose $\beta = \{v_1, \dots, v_n\}$ is an orthonormal basis for V consisting of eigenvectors of T . Suppose further that $\lambda_1, \dots, \lambda_n$ are the corresponding eigenvalues such that $|\lambda_i| = 1$.

Then T is diagonalizable. Hence for the matrix representation A of T ,

$$A = PDP^{-1}$$

where D is a diagonal matrix. In this case, the columns of P are orthonormal, hence $PP^* = I$ and P is unitary.

So $T = UDU^*$ for a unitary operator U and a diagonal operator D . Then

$$\begin{aligned}T^* &= (UDU^*)^* \\ &= UD^*U^* \\ &= U\overline{D}U^* = UDU^*.\end{aligned}$$

Moreover,

$$\begin{aligned}TT^* &= (UDU^*)(UDU^*) \\ &= UDDU^* \\ &= UIU^* = I.\end{aligned}$$

It follows that $T^*T = I$. Hence T is self-adjoint and unitary.

Conversely, suppose T is self-adjoint and unitary. By the Spectral Theorem for self-adjoint operators, there is an orthonormal basis for V of eigenvectors of T . Since T is unitary, $|\lambda_i| = 1$ for each eigenvalue λ_i of T . ■

Proposition

For any linear operator T on an inner product space V , the operator T^*T is positive semidefinite.

Proof: Let $x \in V$. Then

$$\langle T^*T(x), x \rangle = \langle T(x), T(x) \rangle = \|T(x)\|^2 \geq 0.$$

Hence T^*T is positive semidefinite. ■

Corollary

If T is a linear operator on an inner product space, $\sqrt{T^*T}$ exists.

Definition

Singular Values

The **singular values** of a linear operator T are the nonnegative square roots of the eigenvalues of T^*T . That is,

$$\text{SingVal}(T) = \{\sqrt{\lambda_1}, \sqrt{\lambda_2}, \dots, \sqrt{\lambda_n}\} \quad \text{where } \lambda_i \text{ is an eigenvalue of } T^*T.$$

Theorem

Singular Value Decomposition

Let A be an $m \times n$ matrix with real or complex entries. Then $A = UDV^*$ for some unitary matrices U and V and a diagonal matrix D where

$$D = \begin{pmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \end{pmatrix} \quad \text{and } \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$$

whose diagonal entries are the singular values of T .